

Inverse Modeling and Optimization of the Advection–Diffusion Equation

Alaa

June 12, 2026

Abstract

This project presents a computational framework for solving an inverse parameter estimation problem based on the one-dimensional Advection–Diffusion equation. The goal is to estimate unknown physical parameters using numerical simulation combined with optimization algorithms.

1 Introduction

Inverse problems in partial differential equations (PDEs) aim to estimate unknown parameters from observed data.

This project integrates:

- Numerical PDE solver
- Synthetic data generation
- Optimization algorithms (GA and PSO)

2 Mathematical Model

The governing equation is:

$$\frac{\partial u}{\partial t} + v \frac{\partial u}{\partial x} = D \frac{\partial^2 u}{\partial x^2} \quad (1)$$

where $u(x, t)$ is the state variable, v is velocity, and D is diffusion coefficient.

3 Numerical Method

Advection term:

$$\frac{\partial u}{\partial x} \approx \frac{u_i - u_{i-1}}{\Delta x} \quad (2)$$

Diffusion term:

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2} \quad (3)$$

Time update:

$$u^{n+1} = u^n + \Delta t \cdot (\text{advection} + \text{diffusion}) \quad (4)$$

4 Inverse Problem

We minimize:

$$J(v, D) = \|u_{sim}(v, D) - u_{obs}\|_2^2 \quad (5)$$

5 Optimization Methods

5.1 Genetic Algorithm

Population-based search using selection, crossover, and mutation.

5.2 Particle Swarm Optimization

Swarm-based search using global and personal best updates.

6 Experimental Setup

True parameters:

- $v = 0.8$
- $D = 0.05$

7 Results

PSO achieved accurate estimation:

- $v \approx 0.80$
- $D \approx 0.050$

8 Conclusion

The framework successfully solves the inverse PDE problem using optimization methods coupled with numerical simulation.